Kinect-Based Human Motion Analysis

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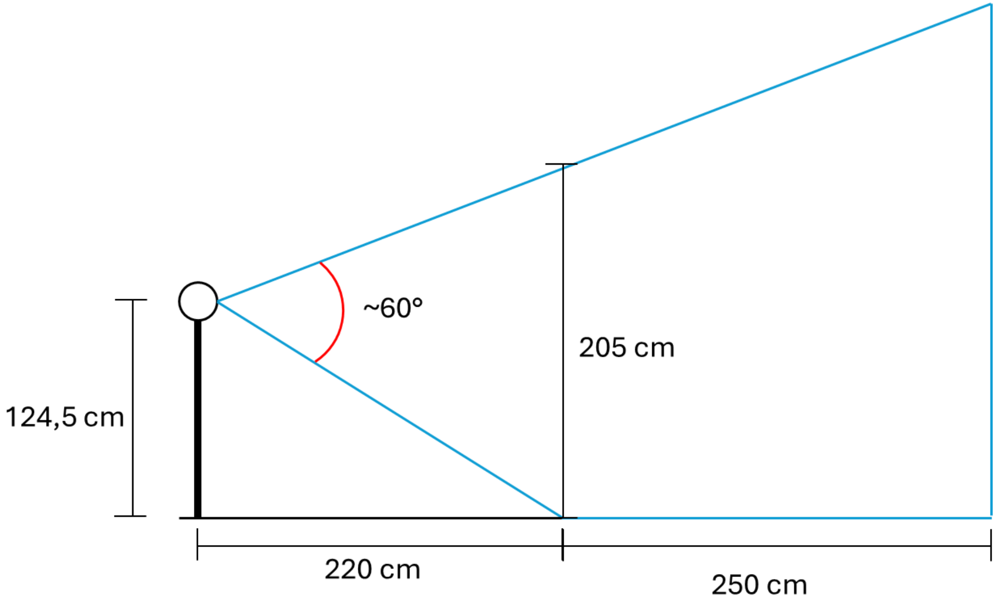
Contents

Experimental Setup

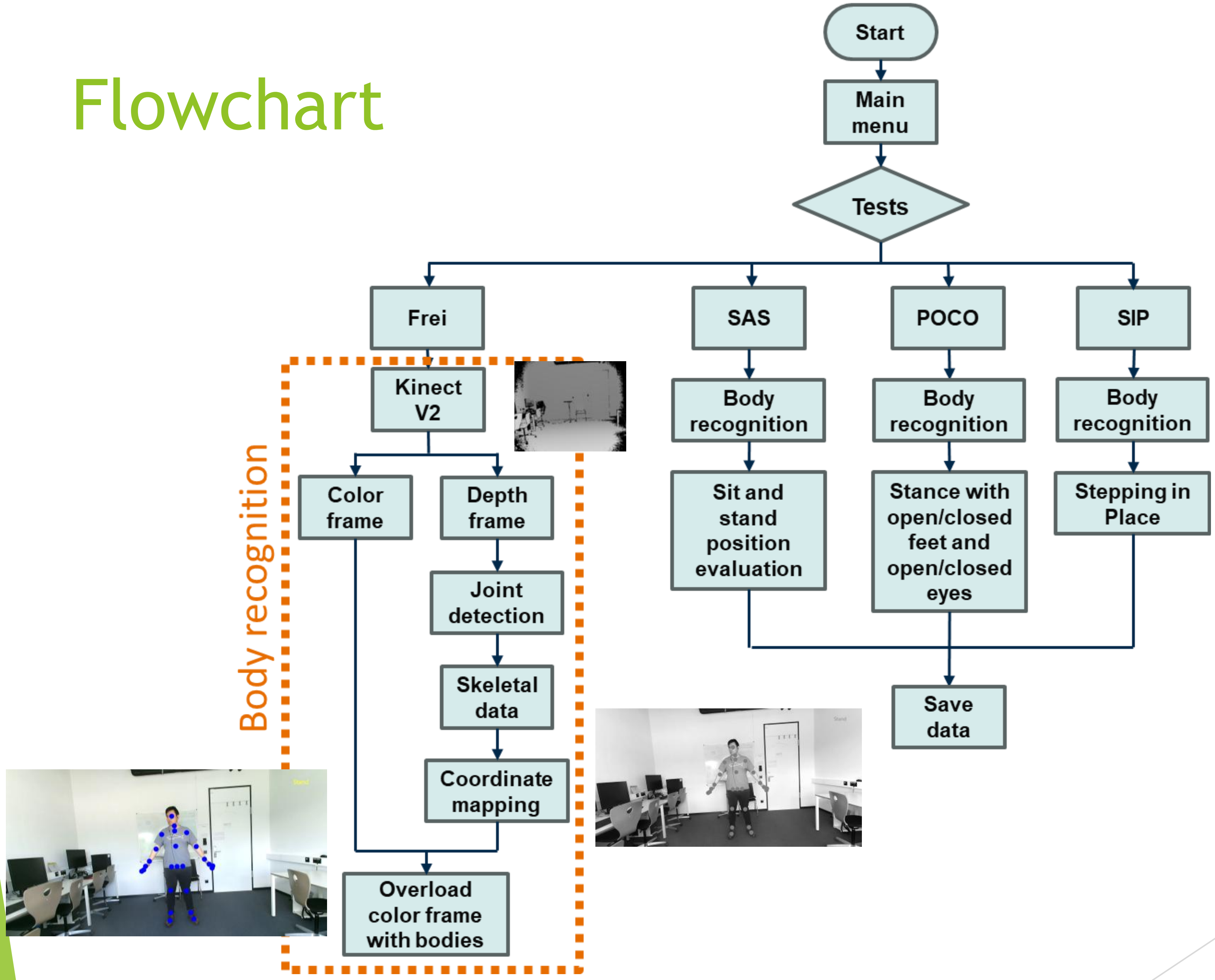
- Materials:
  - Tripod
  - Kinect V2
  - Computer running Windows 11
  - MATLAB

Kinect on a tripod



System Diagram: Implementation

Flowchart


Start
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Body recognition
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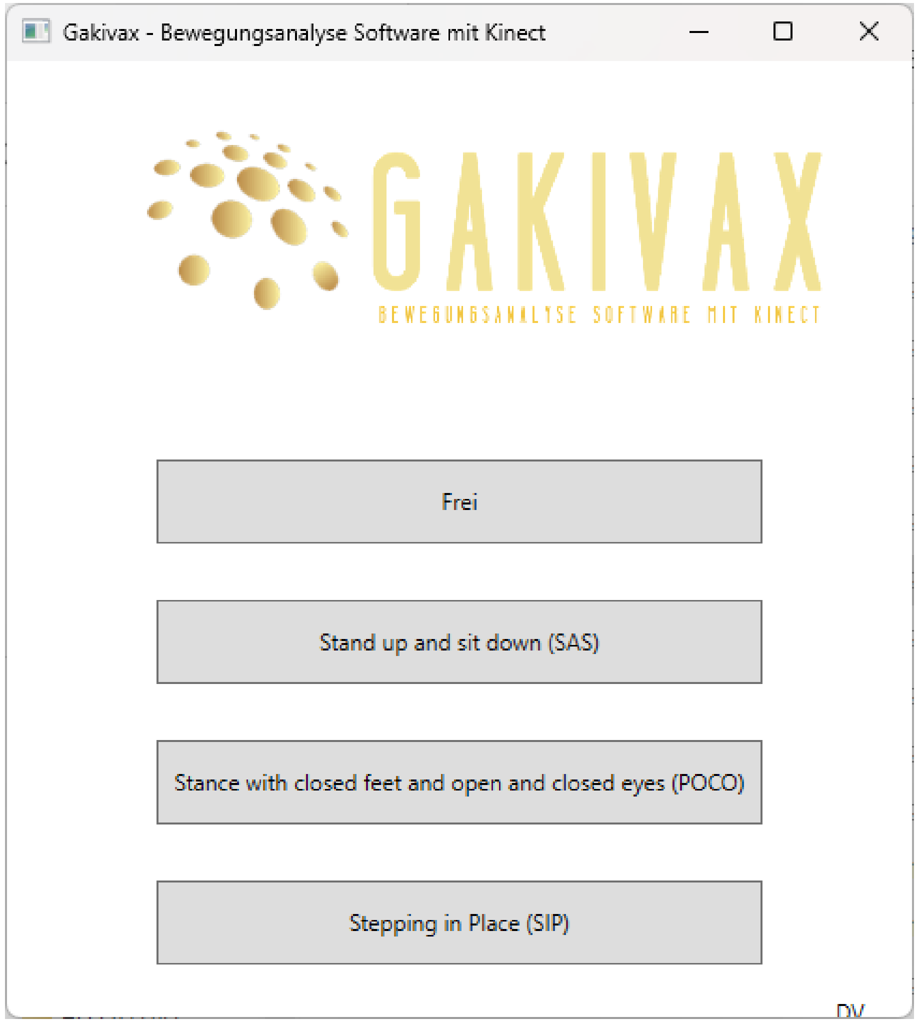
Body recognition
Sit and stand position evaluation
Body recognition
Stance with open/closed feet and open/closed eyes
Body recognition
Stepping in Place
Save data

UI



Main interface

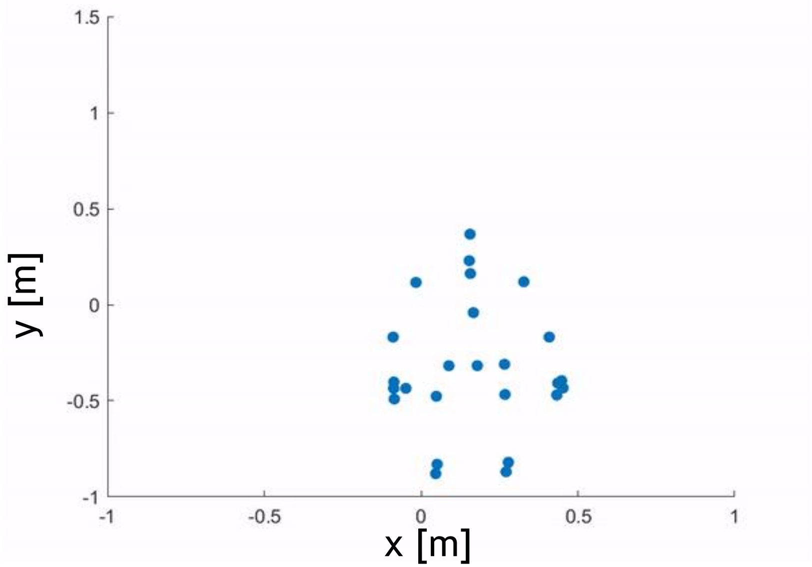
SAS interface

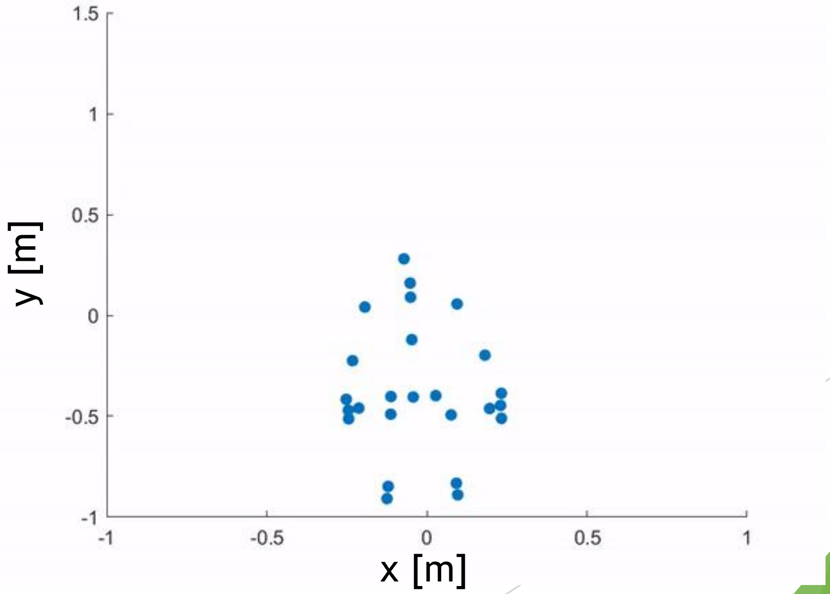
Software Features

- **Frei:** The software detects a person and tracks the 25 joints identified by Kinect
- **Standing and Sitting** (SAS): detects whether the person is sitting or standing
- **Posture with feet together and eyes open and closed** (POCO): detects when the person brings their feet together or apart; the eye status of the person monitoring the test must be confirmed
- **Walking in place** (SIP): detects an upright person to begin data recording

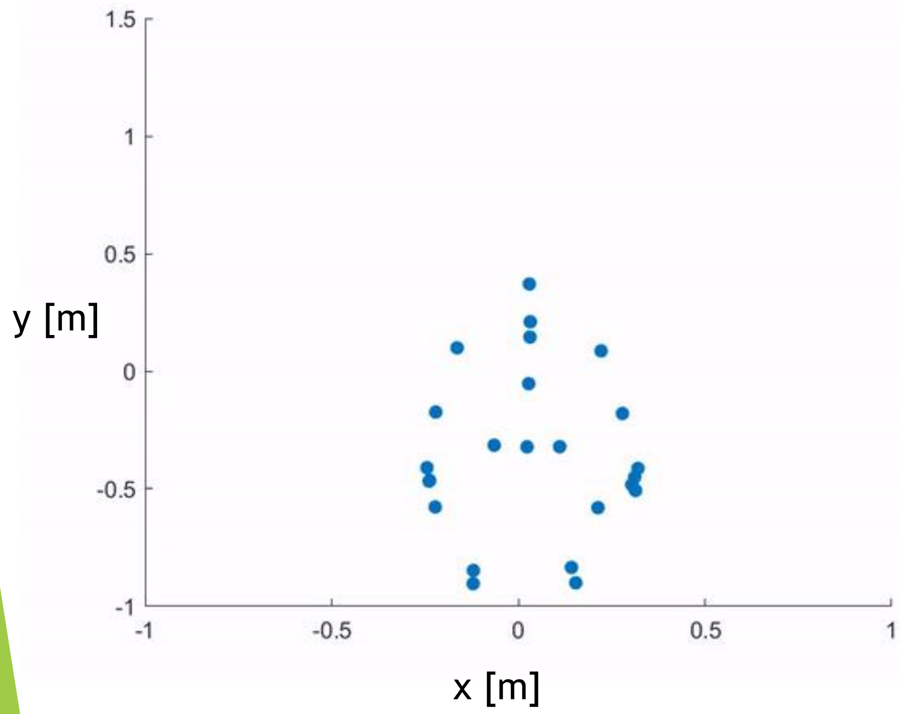
Test Setup

- 5 healthy subjects: 2 women and 3 men (students and staff of TU Ilmenau)
- 3 SAS
  - Standard chair height
  - Comfortable chair height
  - Max. chair height
- POCO
- SIP
- Verify the validity of Kinect for analyzing human movements



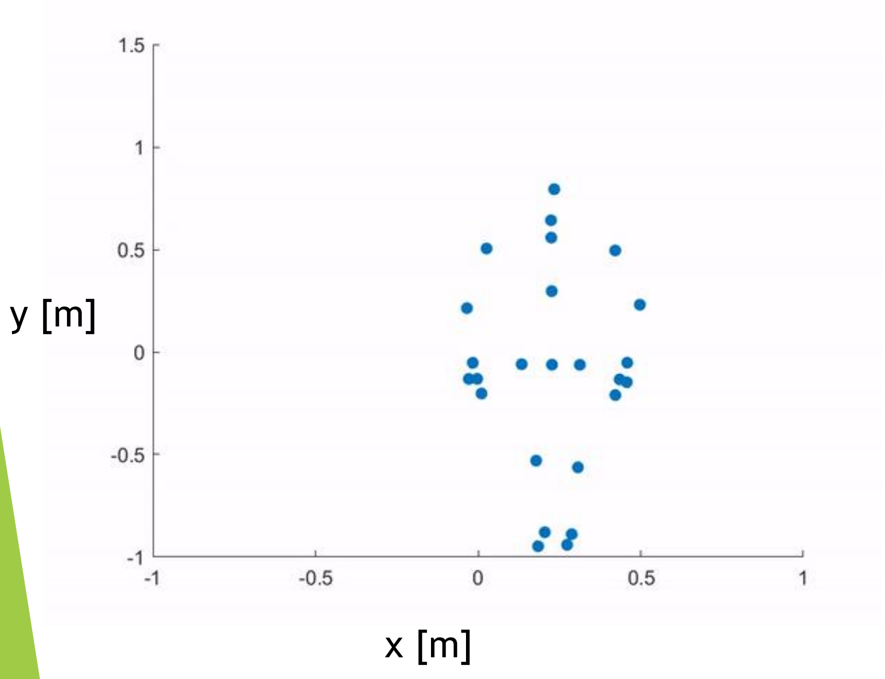


Standing Up and Sitting Down



Test time [s]	55
Number of repetitions:	5
Average displacement [m]:	0.3662
Seat height [cm]	46
↑ - Time [s]	0.9533
↑ Range of motion: medial-lateral [cm]	4.3742
↑ Anterior-posterior range of motion [cm]	43.1813
↑ Range of motion of the right hand [cm]	27.648
↑ Range of motion of the left hand [cm]	27.6949
↓ - Time [s]	0.8667
↓ Range of motion: medial-lateral [cm]	3.3855
↓ Range of motion: Anterior-posterior [cm]	40.5402

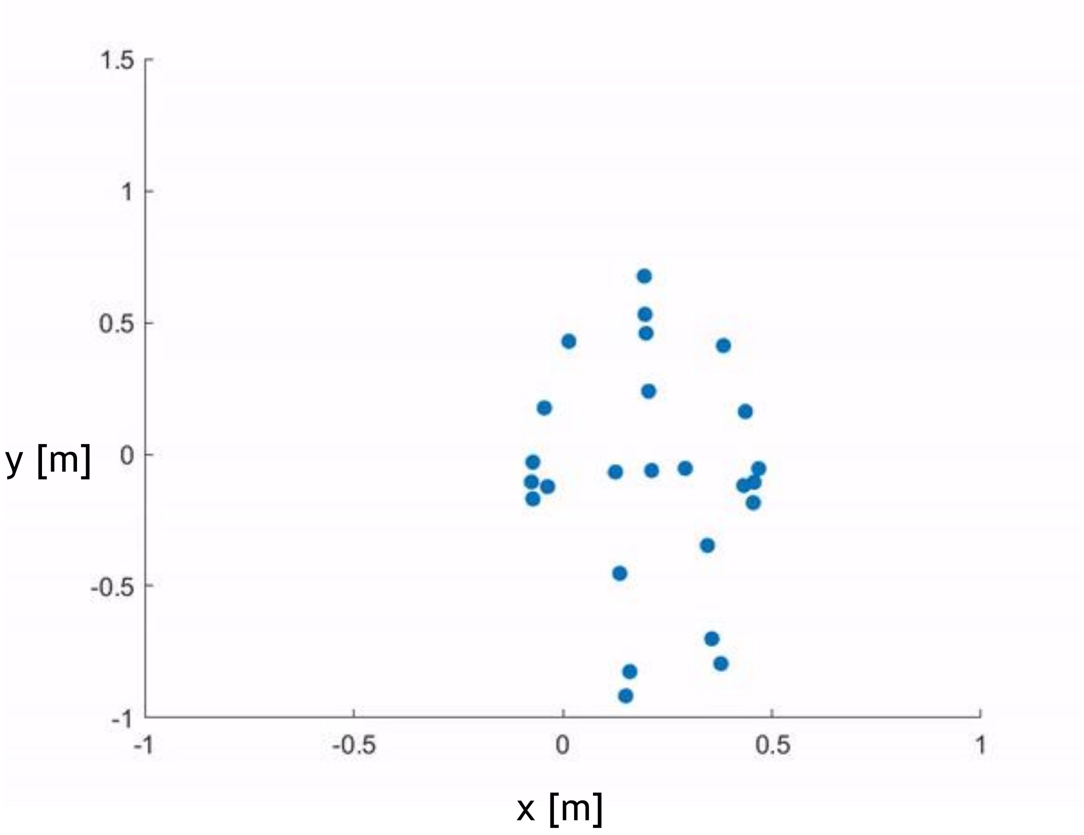
Posture with Feet Together and Eyes Open and Closed



Test time [s]	120
Pitch deflection range [°]	1.0586
Roll deflection range [°]	0.4296
3D deflection range [°]	3.0946
Average pitch fluctuation rate [°/s]	0.0529
Average roll fluctuation rate [°/s]	0.0215
Average 3D fluctuation rate[°/s]	0.1547

Pitch deflection range [°]	0.3751
Roll deflection range [°]	0.4774
3D deflection range [°]	-4.4292
Average pitch fluctuation rate [°/s]	0.0188
Average roll fluctuation rate [°/s]	0.0239
Average 3D fluctuation rate[°/s]	-0.2215

Walking in Place



Test time [s]	125
Range of motion, left knee [cm]	19.81
Range of motion, right knee [cm]	18.63
Cadence left [steps/min]	43.4694
Cadence right [steps/min]	43.299

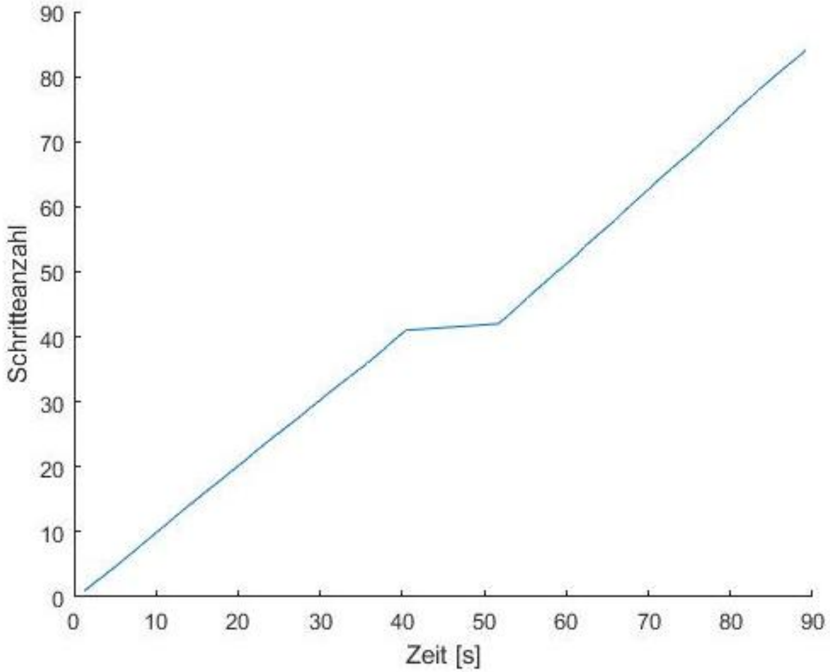
Results

	Gakivax		Kinect (Paper)		Vicon (Paper)	
	Mittelwert	**SA**	Mittelwert	SA	Mittelwert	SA
Aufstehen und Hinsetzen						
Durchschnittliche Verschiebung [m]:	**0.32**	**0.06**	-	-	-	-
↑ - Time [s]	**1.28**	**0.92**	1.31	0.25	1.38	0.25
↑ Range of motion: medial-lateral [cm]	**2.64**	**2.40**	2.42	1.42	2.15	1.16
↑ Anterior-posterior range of motion [cm]	**34.45**	**6.30**	37.02	6.51	37.22	6.41
↑ Range of motion of the right hand [cm]	**25.27**	**9.69**	31.74	6.28	32.99	6.34
↑ Range of motion of the left hand [cm]	**23.53**	**7.95**	34.04	6.96	34.67	7.07
↓ - Time [s]	**1.72**	**1.45**	1.53	0.23	1.59	0.23
↓ Range of motion: medial-lateral [cm]	**1.79**	**2.42**	2.87	1.45	2.37	1.26
↓ Range of motion: Anterior-posterior [cm]	**30.55**	**11.33**	39.67	6.27	39.29	5.97
Haltung mit geschlossenen Füßen und offenen und geschlossenen Augen						
EO Pitch deflection range [°]	**0.75**	**0.30**	1.67	0.76	1.49	0.63
EO Roll deflection range [°]	**0.76**	**0.28**	1.54	0.52	1.35	0.4
EO 3D deflection range [°]	**0.04**	**0.01**	0.17	0.06	0.18	0.06
EO Average pitch fluctuation rate [°/s]	**0.04**	**0.01**	0.21	0.07	0.2	0.06
EC Pitch deflection range [°]	**0.58**	**0.28**	1.67	0.76	1.49	0.63
EC Roll deflection range [°]	**0.81**	**0.31**	1.54	0.52	1.35	0.4
EC 3D deflection range [°]	**0.03**	**0.01**	0.17	0.06	0.18	0.06
EC Average pitch fluctuation rate [°/s]	**0.04**	**0.02**	0.21	0.07	0.2	0.06
Gehen auf der Stelle						
Range of motion, left knee [cm]	**14.34**	**6.34**	4.37	1.06	9.05	1.7
Range of motion, right knee [cm]	**14.09**	**6.47**	4.38	1.02	9.03	1.89
Cadence left [steps/min]	**49.02**	**10.73**	47.08	7.31	47	7.21
Cadence right [steps/min]	**46.56**	**10.08**	47.63	7.18	47.58	7.15

EO EC

Results (2)

- The algorithms used to calculate the individual parameters are not explained in the researched papers
- Development of custom algorithms
- Pseudocode for “marching in place”:



Start
|| Read knee signals (t, x, y, z)
|| Set thresholds for both knees
|| Search for maximum values in the original knee signals
|| Fit a second-degree polynomial trend curve to the found maximum values
|| Use the trend curve to normalize the data and set new thresholds
|| Search for maximum values in the normalized signals of both knees
|| Calculate the parameters
|| Print results
End

Results (3)

- Summary of the results in a report
- Report:
  - key charts for each test
  - calculated numerical values
  - manually done